

Virtual Casino Simulator

9.1.0 Requirements & Validation Report

Control Plane + UX Stabilization Release

Application version	9.1.0	Requirement count	344
Document revision	9.1.0-docs-redesign	Validation status	344 PASS / 0 FAIL
Generated	2026-07-08T17:33:50Z	Format	Landscape PDF for readable diagrams and tables

This redesigned report replaces the cramped diagram pages with a landscape layout, one major diagram per section, cleaner connector routing, compact status summaries, and readable requirement tables.

Scope Control plane, bots, audio, autoplay, roulette wheel, UX stability	Modules 16 separately versioned modules	Coverage 186 requirements with API tests; 76 with browser tests
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1. Executive summary

v9.1.0 is a control-plane and UX stabilization release. It separates bot controllers from game modules, moves global sound and voice configuration into Admin, redesigns the Admin console, centralizes autoplay session state, fixes Roulette wheel result state, and stabilizes game layouts during gameplay.

Status	Count	Meaning
PASS	344	Requirement is implemented and validated by automated checks, static checks, or documented manual evidence.

Requirement distribution by module

Module	Requirements	API-tested	Browser-tested
Admin	22	18	5
Audio	7	3	2
Autoplay	14	5	2
Baccarat	24	20	4
Bingo	24	22	4
Blackjack	30	10	2
Bots	8	3	0
Core	20	4	3
Documentation	15	3	0
Keno	22	11	14
Ledger	25	15	0
Logging	15	5	1
Roulette	56	42	12
Slots	26	19	7
Testing	24	3	15
Tests	6	3	2
UX	6	0	3

2. Module revisions

Each module has its own revision so future work can update one module without forcing a full casino-wide version bump.

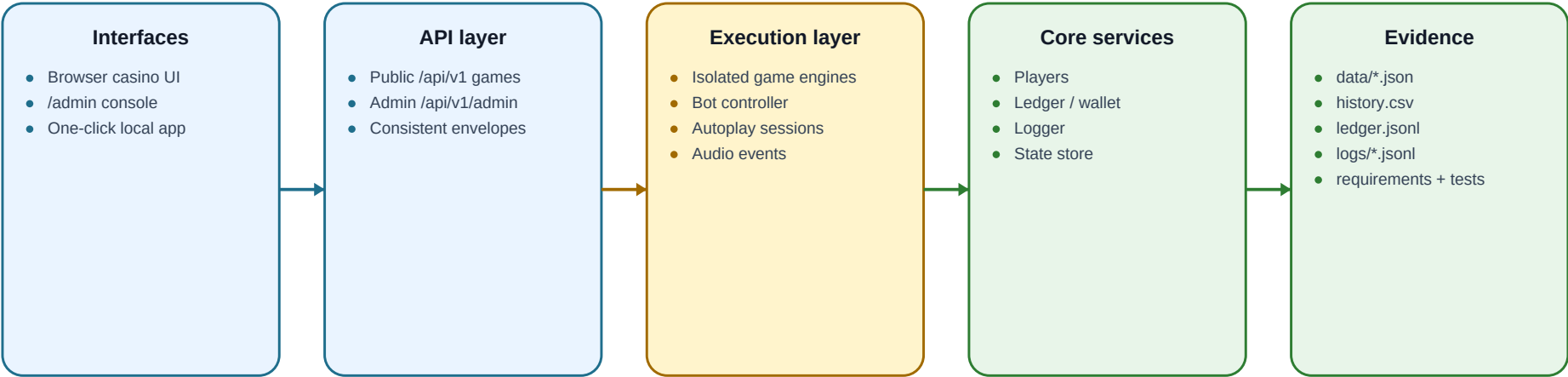
Module	Revision	Notes
Core	9.1.0	App shell, API router, static asset server.
Ledger	9.0.1	Shared wallet audit trail and money movement.
Players	9.0.1	Human and bot-owned player records.
Bot Controller	1.0.0	Standalone bot profiles, capabilities, and strategies.
Autoplay Controller	1.1.0	Server-registered autoplay sessions and stop contract.
Audio Voice	9.1.0	Global audio/voice settings under Admin.
Logging	9.1.0	App, error, client, and telemetry logs.
Roulette	9.1.0	Wheel state, table bets, bots, stats, autoplay template.
Slots	9.0.1	Reel-strip slots and autoplay compatibility.
Blackjack	9.0.1	Manual blackjack with rule and action APIs.
Baccarat	9.0.1	Punto Banco shoe and bot betting.
Keno	9.0.1	1-20 spot tickets and draw logic.
Bingo	9.0.1	75-ball American Bingo with bot cards.
Admin	1.1.0	Control-plane console.
Tests	1.1.0	API, rule, and browser harness.
Docs	1.1.0	Requirement registry and validation reports.

3. Architecture diagrams

The diagrams are separated into focused views. Connectors are routed horizontally or through explicit buses to avoid the line-crossing issues in the previous document.

3.1 Control-plane architecture

CONTROL-PLANE DATA FLOW

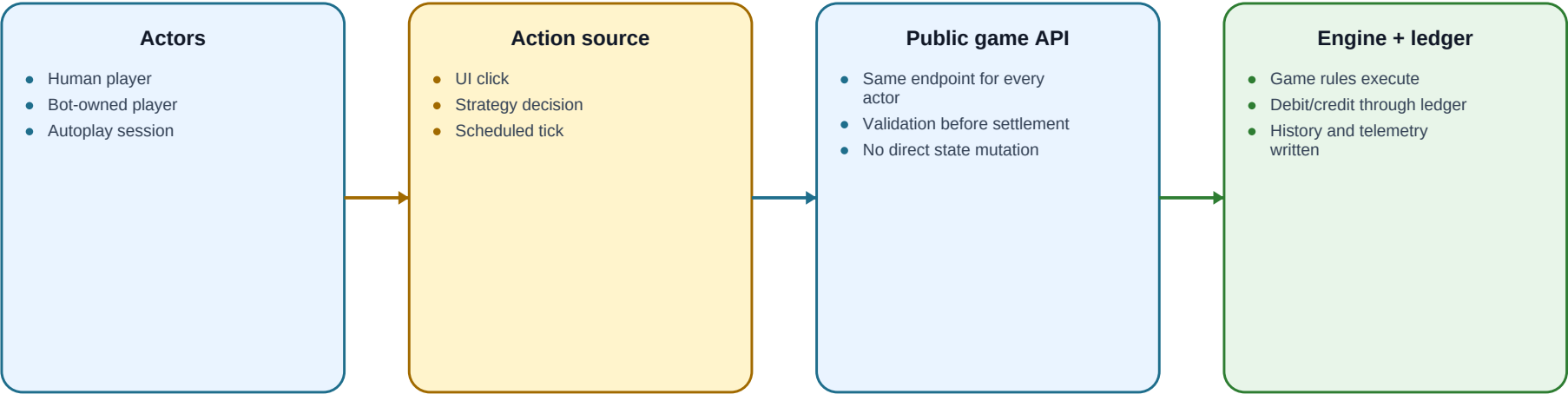


Design rule: games stay isolated; shared services own money, state, logs, settings, and validation evidence.

Casino UI and Admin are separate surfaces. Both use versioned APIs. Game engines remain isolated; shared money, state, logs, and settings are handled by core services.

3.2 Bot separation

BOT CONTROL MODEL

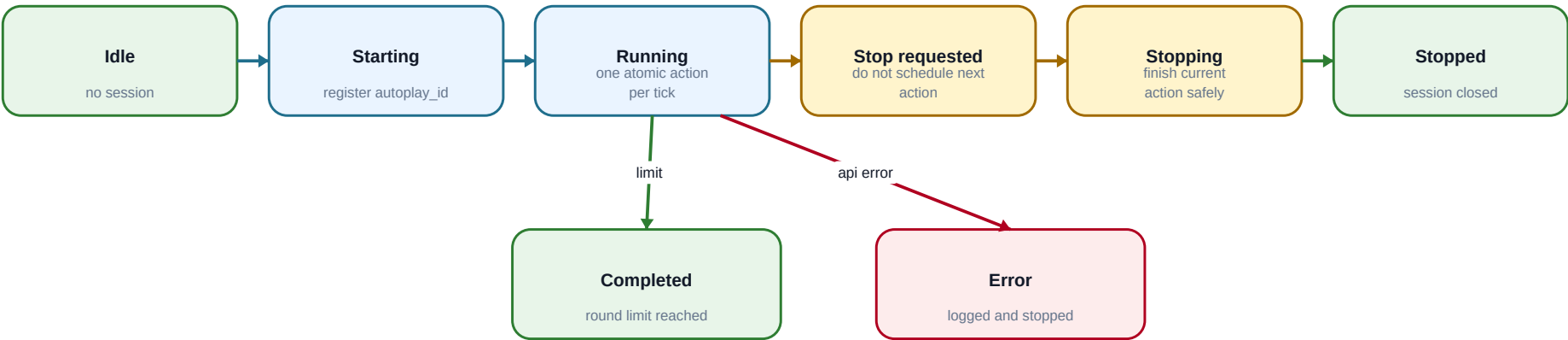


Rule: a bot is a controller for a player account. A game exposes capabilities but does not own bot strategy configuration.

Bots are controllers for player accounts, not objects embedded inside games. Bot strategies produce legal actions, then submit them through the same public APIs used by human and autoplay actions.

3.3 Autoplay state machine

AUTOPLAY SESSION STATES

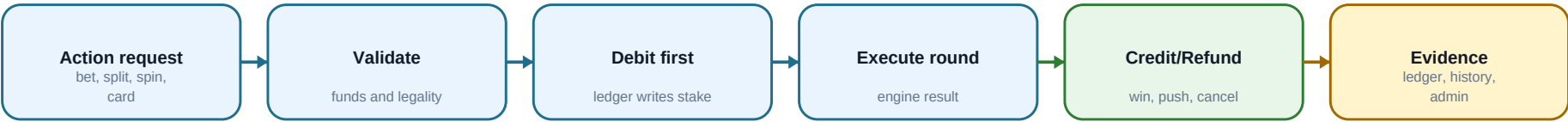


Stop is cooperative: current atomic action settles safely; no new round, draw, coup, card, or spin starts after stop is requested.

Autoplay is centrally registered. Stop is cooperative and deterministic: the current atomic action is allowed to settle safely, then no next action is scheduled.

3.4 Money lifecycle

MONEY MOVEMENT CONTRACT



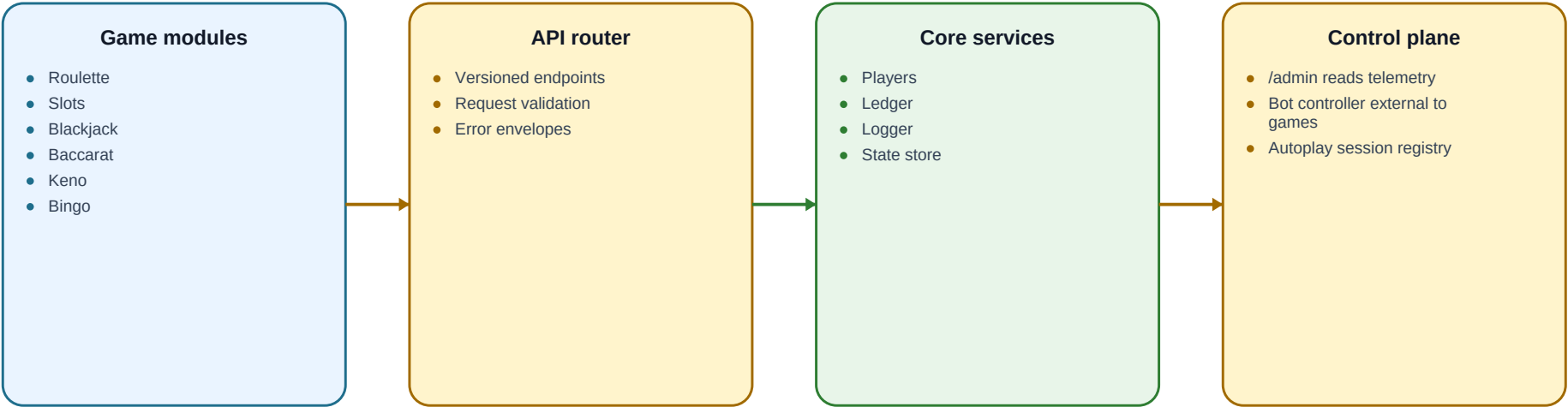
Game	Money lifecycle rule
Roulette	Place chip debits immediately; clear-before-spin refunds; winning settlement credits total return.
Blackjack	Deal debits initial bet; double/split/insurance each use separate debit events.
Keno/Bingo	Ticket/card purchase debits first; draw/call resolution credits winnings or records forfeiture.

The ledger is the source of truth for every debit, credit, refund, push, split, double, ticket purchase, card purchase, and settlement.

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3.5 Runtime isolation

RUNTIME ISOLATION



Rule: game modules do not import each other. Shared concerns flow through core services and versioned APIs.

Game modules route through shared core services but do not import each other. This prevents one game state or variable from breaking another game page.

4. API surface summary

The API is versioned under /api/v1. Human clicks, autoplay, and bot controllers use the same public game endpoints, while Admin uses control-plane endpoints.

Area	Representative endpoints	Purpose
Casino	/api/v1/casino/state, /api/v1/casino/reset, /api/v1/casino/history	Application state, reset, and global history.
Players	/api/v1/players, /api/v1/players/{id}, /api/v1/players/{id}/add-money	Player records and fake-money balance management.
Roulette	/api/v1/games/roulette/state, /bets, /spin, /rebet, /stats	Bet placement, spin settlement, stats, and templates.
Slots	/api/v1/games/slots/state, /spin	Reel stop generation and payout settlement.
Blackjack	/api/v1/games/blackjack/rounds, /hit, /stand, /double, /split, /surrender, /insurance	Manual blackjack round lifecycle and separate action debits.
Baccarat	/api/v1/games/baccarat/bets, /deal, /state	Punto Banco betting, shoe, deal, and settlement.
Keno	/api/v1/games/keno/tickets, /draw, /state	Ticket purchase, cancellation, and 20-number draw.
Bingo	/api/v1/games/bingo/cards, /call, /auto-step, /reset	Card purchase, ball calls, and stepwise autoplay.
Bots	/api/v1/bots, /api/v1/bots/capabilities, /api/v1/games/{game_id}/eligible-bots	Bot profiles, capabilities, and strategy routing.
Autoplay	/api/v1/autoplay/start, /stop, /tick, /complete, /stop-all, /sessions	Central session lifecycle and stop contract.
Admin	/api/v1/admin/dashboard, /logs, /ledger, /history, /game-states, /requirements, /audio-settings	Telemetry, settings, and operational visibility.

5. New or changed requirements in v9.1.0

57 requirements were introduced or changed in this release. They are grouped around bot separation, global audio control, Admin redesign, autoplay, Roulette wheel state, layout stability, and test harness improvements.

ID	Module	Requirement	Status	Tests
BOT-001	Bots	Bots are represented as controllers for player accounts, not embedded game objects.	PASS	API-CONTROL-001
BOT-002	Bots	Game modules must not import bot strategy modules.	PASS	
BOT-003	Bots	A bot appears for a game only when it has a compatible strategy.	PASS	API-CONTROL-001
BOT-004	Bots	Bot actions use the same public game APIs and engine validation paths as human actions.	PASS	
BOT-005	Bots	Bot money movement uses the shared ledger system.	PASS	API-CONTROL-001
BOT-006	Bots	Bot strategy assignment is visible and editable from Admin.	PASS	
BOT-007	Bots	Bot actions are logged with bot_id, player_id, game_id, round context, and strategy_id where applicable.	PASS	
BOT-008	Bots	Unsupported games hide bot controllers rather than showing incompatible bot settings.	PASS	
AUDIO-001	Audio	Sound configuration is global and not owned by any game page.	PASS	API-CONTROL-001
AUDIO-002	Audio	Full Sound and Voice settings live under /admin.	PASS	API-CONTROL-001, BR-AUDIO-001
AUDIO-003	Audio	Game pages may trigger sound events but do not render full sound settings panels.	PASS	
AUDIO-004	Audio	Audio settings persist in data/settings/audio.json.	PASS	API-CONTROL-001
AUDIO-005	Audio	Voice preview uses the currently selected settings.	PASS	BR-AUDIO-001
AUDIO-006	Audio	Master mute stops new sound effects and voice announcements.	PASS	
AUDIO-007	Audio	Per-game voice announcements are individually configurable.	PASS	
AUTO-001	Autoplay	Autoplay is centrally controlled rather than implemented as unrelated game loops.	PASS	API-CONTROL-001
AUTO-002	Autoplay	Every autoplay run has an autoplay_id.	PASS	API-CONTROL-001
AUTO-003	Autoplay	Stop prevents any new round or action from starting.	PASS	API-CONTROL-001, BR-AUTO-ROU-001
AUTO-004	Autoplay	Stop during an atomic action completes that action safely and schedules no follow-up action.	PASS	
AUTO-005	Autoplay	Autoplay speed consistently affects inter-round delay.	PASS	
AUTO-006	Autoplay	Autoplay logs start, stop, tick, completion, and error status through the server session store.	PASS	
AUTO-007	Autoplay	Admin shows active and recent autoplay sessions.	PASS	API-CONTROL-001
AUTO-008	Autoplay	Admin Stop All requests stop for all server-registered autoplay sessions.	PASS	API-CONTROL-001
AUTO-009	Autoplay	Roulette autoplay repeats the saved bet template.	PASS	BR-AUTO-ROU-001
AUTO-010	Autoplay	Slots autoplay repeats current spin parameters.	PASS	
AUTO-011	Autoplay	Baccarat autoplay repeats selected standing wagers.	PASS	
AUTO-012	Autoplay	Keno autoplay repeats the selected ticket.	PASS	
AUTO-013	Autoplay	Bingo autoplay uses stepwise ball calls so Stop is honored between calls.	PASS	

ID	Module	Requirement	Status	Tests
AUTO-014	Autoplay	Blackjack autoplay remains disabled unless an explicit strategy controller is later enabled.	PASS	
ADMIN-013	Admin	Admin console uses a dedicated sidebar/topbar control-plane layout.	PASS	API-ADMIN-001, BR-ADMIN-001
ADMIN-014	Admin	Admin Dashboard shows version, players, bots, autoplay, errors, and requirement summary.	PASS	API-ADMIN-001
ADMIN-015	Admin	Admin Players & Bots tab shows player balances and bot strategy assignments.	PASS	
ADMIN-016	Admin	Admin Ledger tab shows transaction audit rows.	PASS	API-ADMIN-001
ADMIN-017	Admin	Admin Telemetry tab shows app, error, and client logs.	PASS	
ADMIN-018	Admin	Admin Game States tab shows isolated game state files.	PASS	
ADMIN-019	Admin	Admin Audio & Voice tab stores global audio settings.	PASS	API-ADMIN-001, BR-ADMIN-001
ADMIN-020	Admin	Admin Autoplay tab shows sessions and Stop All.	PASS	API-ADMIN-001
ADMIN-021	Admin	Admin Requirements tab shows requirement coverage.	PASS	API-ADMIN-001
ADMIN-022	Admin	Admin Tests tab shows latest test results.	PASS	API-ADMIN-001
ROU-051	Roulette	Roulette wheel must not default to a fake zero result when no spin has occurred.	PASS	
ROU-052	Roulette	Roulette wheel shows the latest actual spin result after settlement.	PASS	BR-ROU-001
ROU-053	Roulette	Roulette wheel selected pocket and backend spin result must match.	PASS	BR-ROU-001
ROU-054	Roulette	Roulette spin animation starts before settlement display and ends on the result state.	PASS	
ROU-055	Roulette	Roulette ball indicator uses the selected pocket when a result exists.	PASS	
ROU-056	Roulette	Roulette sound settings are not rendered on the Roulette page.	PASS	
UX-001	UX	Game stages reserve fixed visual areas during normal gameplay.	PASS	BR-ROU-001
UX-002	UX	Action rails remain stable while actions execute.	PASS	BR-ROU-001
UX-003	UX	Result messages render in fixed-height regions.	PASS	BR-ROU-001
UX-004	UX	Long history, stats, paytables, and logs use internal scroll areas.	PASS	
UX-005	UX	Autoplay status changes do not resize the main game stage.	PASS	
UX-006	UX	Animations prefer transform/opacity and avoid layout-changing motion.	PASS	
TEST-025	Tests	Browser tests use stable data-testid selectors for autoplay and admin controls.	PASS	BR-AUTO-ROU-001, BR-AUDIO-001
TEST-026	Tests	Browser tests verify Roulette autoplay stop behavior.	PASS	BR-AUTO-ROU-001, BR-AUDIO-001
TEST-027	Tests	API tests verify bot controller endpoints.	PASS	API-CONTROL-001
TEST-028	Tests	API tests verify persisted audio settings.	PASS	API-CONTROL-001
TEST-029	Tests	API tests verify server-registered autoplay session lifecycle.	PASS	API-CONTROL-001
TEST-030	Tests	Test results remain visible in Admin.	PASS	

6. Validation evidence

The packaging validation combines Python compile checks, rule checks, API checks, JavaScript syntax checks, and a Playwright browser harness for local execution.

Check set	Command	Expected result
Python compile	python -m py_compile run.py \$(find casino -name *.py) verify_rules.py tests/run_tests.py	No syntax errors.
Rule checks	python verify_rules.py	32 passed.
API checks	python tests/run_tests.py --api	9 passed, 0 failed.
JavaScript static checks	node --check web/app.js web/core/*.js web/games/*.js web/admin.js	No syntax errors.
Browser checks	python3 -m playwright install chromium; python3 tests/run_tests.py --browser	Runs locally with screenshots and logs on failure.

Automated coverage by requirement

Coverage type	Requirement count	Notes
API mapped	186	Requirement has one or more API or rule test identifiers.
Browser mapped	76	Requirement has one or more Playwright browser test identifiers.
Documented/static/manual evidence	98	Requirement is validated by code review, static checks, or documented packaging evidence.

7. Release notes summary

Bot controller separation

Added a standalone bot controller layer with bot profiles, capabilities, strategies, and shared API/ledger paths. Games no longer own bot configuration.

Global audio and voice

Moved full audio and voice settings to Admin. Roulette and other game pages only trigger sound events.

Admin redesign

Rebuilt Admin as a sidebar/topbar control plane with dashboard, players/bots, ledger, history, telemetry, game states, audio, autoplay, requirements, tests, and system tabs.

Autoplay controller

Added server-registered autoplay sessions with autoplay_id, stop request semantics, stepwise Bingo calls, and Admin Stop All.

Roulette wheel and UX

Roulette no longer defaults to fake zero state; the wheel highlights the latest actual result and layout stability styles reduce screen movement.

Tests and docs

Requirement registry increased to 344 requirements with updated API and browser harnesses. This document redesign improves readability and diagram quality.

8. Full requirement registry

The following tables contain every tracked requirement in the v9.1.0 registry. Status values are shown exactly as recorded in docs/requirements.json.

8.1 Admin

ID	Requirement	Status	API tests	Browser tests
ADMIN-001	Admin console is available at /admin.	PASS	API-ADMIN-001	BR-ADMIN-001
ADMIN-002	Admin console is unauthenticated for local use.	PASS	API-ADMIN-001	
ADMIN-003	Admin overview shows version and requirement counts.	PASS	API-ADMIN-001	BR-ADMIN-001
ADMIN-004	Admin modules tab shows module revision numbers.	PASS	API-ADMIN-001	
ADMIN-005	Admin players tab shows player balances.	PASS	API-ADMIN-001	
ADMIN-006	Admin ledger tab shows recent ledger events.	PASS	API-ADMIN-001	
ADMIN-007	Admin history tab shows recent history rows.	PASS	API-ADMIN-001	
ADMIN-008	Admin logs tab shows app/error/client logs.	PASS	API-ADMIN-001	
ADMIN-009	Admin game-states tab shows isolated game states.	PASS	API-ADMIN-001	
ADMIN-010	Admin requirements tab shows requirement coverage.	PASS	API-ADMIN-001	
ADMIN-011	Admin test-results tab shows latest test results.	PASS	API-ADMIN-001	
ADMIN-012	Admin has a return-to-casino control.	PASS		BR-ADMIN-001
ADMIN-013	Admin console uses a dedicated sidebar/topbar control-plane layout.	PASS	API-ADMIN-001	BR-ADMIN-001
ADMIN-014	Admin Dashboard shows version, players, bots, autoplay, errors, and requirement summary.	PASS	API-ADMIN-001	
ADMIN-015	Admin Players & Bots tab shows player balances and bot strategy assignments.	PASS		
ADMIN-016	Admin Ledger tab shows transaction audit rows.	PASS	API-ADMIN-001	
ADMIN-017	Admin Telemetry tab shows app, error, and client logs.	PASS		
ADMIN-018	Admin Game States tab shows isolated game state files.	PASS		
ADMIN-019	Admin Audio & Voice tab stores global audio settings.	PASS	API-ADMIN-001	BR-ADMIN-001
ADMIN-020	Admin Autoplay tab shows sessions and Stop All.	PASS	API-ADMIN-001	
ADMIN-021	Admin Requirements tab shows requirement coverage.	PASS	API-ADMIN-001	
ADMIN-022	Admin Tests tab shows latest test results.	PASS	API-ADMIN-001	

8.2 Audio

ID	Requirement	Status	API tests	Browser tests
AUDIO-001	Sound configuration is global and not owned by any game page.	PASS	API-CONTROL-001	
AUDIO-002	Full Sound and Voice settings live under /admin.	PASS	API-CONTROL-001	BR-AUDIO-001
AUDIO-003	Game pages may trigger sound events but do not render full sound settings panels.	PASS		
AUDIO-004	Audio settings persist in data/settings/audio.json.	PASS	API-CONTROL-001	
AUDIO-005	Voice preview uses the currently selected settings.	PASS		BR-AUDIO-001

ID	Requirement	Status	API tests	Browser tests
AUDIO-006	Master mute stops new sound effects and voice announcements.	PASS		
AUDIO-007	Per-game voice announcements are individually configurable.	PASS		

8.3 Autoplay

ID	Requirement	Status	API tests	Browser tests
AUTO-001	Autoplay is centrally controlled rather than implemented as unrelated game loops.	PASS	API-CONTROL-001	
AUTO-002	Every autoplay run has an autoplay_id.	PASS	API-CONTROL-001	
AUTO-003	Stop prevents any new round or action from starting.	PASS	API-CONTROL-001	BR-AUTO-ROU-001
AUTO-004	Stop during an atomic action completes that action safely and schedules no follow-up action.	PASS		
AUTO-005	Autoplay speed consistently affects inter-round delay.	PASS		
AUTO-006	Autoplay logs start, stop, tick, completion, and error status through the server session store.	PASS		
AUTO-007	Admin shows active and recent autoplay sessions.	PASS	API-CONTROL-001	
AUTO-008	Admin Stop All requests stop for all server-registered autoplay sessions.	PASS	API-CONTROL-001	
AUTO-009	Roulette autoplay repeats the saved bet template.	PASS		BR-AUTO-ROU-001
AUTO-010	Slots autoplay repeats current spin parameters.	PASS		
AUTO-011	Baccarat autoplay repeats selected standing wagers.	PASS		
AUTO-012	Keno autoplay repeats the selected ticket.	PASS		
AUTO-013	Bingo autoplay uses stepwise ball calls so Stop is honored between calls.	PASS		
AUTO-014	Blackjack autoplay remains disabled unless an explicit strategy controller is later enabled.	PASS		

8.4 Baccarat

ID	Requirement	Status	API tests	Browser tests
BAC-001	Baccarat implements Punto Banco Player/Banker/Tie betting.	PASS	API-BAC-001	
BAC-002	Baccarat uses configurable 6 or 8 deck shoe.	PASS	API-BAC-001	
BAC-003	Baccarat uses persistent shoe state.	PASS	API-BAC-001	
BAC-004	Baccarat burn-card procedure runs when shoe is created.	PASS	API-BAC-001	
BAC-005	Baccarat cut-card reshuffle threshold is supported.	PASS	API-BAC-001	
BAC-006	Baccarat deals in Player, Banker, Player, Banker order.	PASS	API-BAC-001	
BAC-007	Aces count as one.	PASS	API-BAC-001	
BAC-008	Tens and face cards count as zero.	PASS	API-BAC-001	
BAC-009	Hand totals are modulo 10.	PASS	API-BAC-001	
BAC-010	Natural 8 or 9 stops drawing.	PASS	API-BAC-001	
BAC-011	Player draws on 0-5 and stands on 6-7.	PASS	API-BAC-001	
BAC-012	Banker draw tableau is implemented.	PASS	API-BAC-001	
BAC-013	Player bet pays 1:1.	PASS	API-BAC-001	

ID	Requirement	Status	API tests	Browser tests
BAC-014	Banker bet pays with 5 percent commission by default.	PASS	API-BAC-001	
BAC-015	Tie bet pays configurable 8:1 default.	PASS	API-BAC-001	
BAC-016	Player/Banker bets push on tie.	PASS	API-BAC-001	
BAC-017	Baccarat bets debit immediately.	PASS	API-BAC-001	
BAC-018	Baccarat bet cancellation refunds before deal.	PASS	API-BAC-001	
BAC-019	Baccarat bots can place strategy bets.	PASS	API-BAC-001	
BAC-020	Baccarat UI shows cards, totals, and winner.	PASS		BR-BAC-001
BAC-021	Baccarat UI shows road history.	PASS		BR-BAC-001
BAC-022	Baccarat UI shows shoe and burn info.	PASS		BR-BAC-001
BAC-023	Baccarat auto play repeats selected bet.	PASS		BR-BAC-001
BAC-024	Baccarat writes history rows.	PASS	API-BAC-001	

8.5 Bingo

ID	Requirement	Status	API tests	Browser tests
BINGO-001	Bingo uses 75-ball American rules.	PASS	API-BINGO-001	
BINGO-002	Bingo card has B column 1-15.	PASS	API-BINGO-001	
BINGO-003	Bingo card has I column 16-30.	PASS	API-BINGO-001	
BINGO-004	Bingo card has N column 31-45.	PASS	API-BINGO-001	
BINGO-005	Bingo card has G column 46-60.	PASS	API-BINGO-001	
BINGO-006	Bingo card has O column 61-75.	PASS	API-BINGO-001	
BINGO-007	Bingo card has free center space.	PASS	API-BINGO-001	
BINGO-008	Bingo supports any-line pattern.	PASS	API-BINGO-001	
BINGO-009	Bingo supports four-corners pattern.	PASS	API-BINGO-001	
BINGO-010	Bingo supports postage-stamp pattern.	PASS	API-BINGO-001	
BINGO-011	Bingo supports blackout pattern.	PASS	API-BINGO-001	
BINGO-012	Bingo card purchase debits immediately.	PASS	API-BINGO-001	
BINGO-013	Bingo reset before called balls refunds cards.	PASS	API-BINGO-001	
BINGO-014	Bingo reset after called balls logs abandoned session.	PASS	API-BINGO-001	
BINGO-015	Bingo calls unique balls.	PASS	API-BINGO-001	
BINGO-016	Bingo called balls use B/I/N/G/O labels.	PASS	API-BINGO-001	
BINGO-017	Bingo marks called cells.	PASS		BR-BINGO-001
BINGO-018	Bingo highlights winning pattern.	PASS		BR-BINGO-001
BINGO-019	Bingo supports bot cards.	PASS	API-BINGO-001	
BINGO-020	Bingo awards payout to the winning card.	PASS	API-BINGO-001	
BINGO-021	Bingo auto play calls until a winner.	PASS	API-BINGO-001	BR-BINGO-001

ID	Requirement	Status	API tests	Browser tests
BINGO-022	Bingo UI shows cards in play.	PASS	API-BINGO-001	BR-BINGO-001
BINGO-023	Bingo writes history rows.	PASS	API-BINGO-001	
BINGO-024	Bingo remains fake-money only.	PASS	API-BINGO-001	

8.6 Blackjack

ID	Requirement	Status	API tests	Browser tests
BJ-001	Blackjack deals two player cards and two dealer cards.	PASS	API-BJ-001	
BJ-002	Dealer hole card is hidden in public state.	PASS		
BJ-003	Aces can count as 1 or 11.	PASS		
BJ-004	Soft totals are computed correctly for multi-ace hands.	PASS		
BJ-005	Natural blackjack pays configured blackjack payout.	PASS		
BJ-006	Push returns the stake.	PASS		
BJ-007	Dealer stands or hits soft 17 according to setting.	PASS		
BJ-008	Player can hit during active hand.	PASS		
BJ-009	Player can stand during active hand.	PASS		
BJ-010	Double down is available when legal.	PASS	API-BJ-001	
BJ-011	Double down debits an additional wager.	PASS	API-BJ-001	
BJ-012	Double down deals exactly one card and stands.	PASS		
BJ-013	Split is available for equal blackjack values.	PASS	API-BJ-001	
BJ-014	Split debits an additional wager.	PASS	API-BJ-001	
BJ-015	Resplit is limited by max_split_hands.	PASS		
BJ-016	Double-after-split setting is enforced.	PASS		
BJ-017	Split aces one-card rule is enforced.	PASS		
BJ-018	Late surrender setting is enforced.	PASS		
BJ-019	Surrender credits half the wager.	PASS		
BJ-020	Insurance is available only against dealer Ace.	PASS	API-BJ-001	
BJ-021	Insurance cannot be bought more than once.	PASS	API-BJ-001	
BJ-022	Insurance maximum is half original wager.	PASS		
BJ-023	Even money is available with player blackjack against dealer Ace.	PASS		
BJ-024	Blackjack table rules cannot change during active rounds.	PASS	API-BJ-001	
BJ-025	A player cannot start multiple active blackjack rounds.	PASS	API-BJ-001	
BJ-026	Blackjack uses a persistent shoe.	PASS		
BJ-027	Blackjack state endpoint exposes shoe count.	PASS	API-BJ-001	
BJ-028	Blackjack UI renders dealer and player hands.	PASS		BR-BJ-001
BJ-029	Blackjack UI exposes action buttons.	PASS		BR-BJ-001

ID	Requirement	Status	API tests	Browser tests
BJ-030	Blackjack writes history rows on settlement.	PASS		

8.7 Bots

ID	Requirement	Status	API tests	Browser tests
BOT-001	Bots are represented as controllers for player accounts, not embedded game objects.	PASS	API-CONTROL-001	
BOT-002	Game modules must not import bot strategy modules.	PASS		
BOT-003	A bot appears for a game only when it has a compatible strategy.	PASS	API-CONTROL-001	
BOT-004	Bot actions use the same public game APIs and engine validation paths as human actions.	PASS		
BOT-005	Bot money movement uses the shared ledger system.	PASS	API-CONTROL-001	
BOT-006	Bot strategy assignment is visible and editable from Admin.	PASS		
BOT-007	Bot actions are logged with bot_id, player_id, game_id, round context, and strategy_id where applicable.	PASS		
BOT-008	Unsupported games hide bot controllers rather than showing incompatible bot settings.	PASS		

8.8 Core

ID	Requirement	Status	API tests	Browser tests
CORE-001	Application runs locally in a browser from one-click launchers.	PASS		
CORE-002	Windows launcher starts the local Python server.	PASS		
CORE-003	macOS launcher starts the local Python server.	PASS		
CORE-004	Application remains fake-money only with no real payment flows.	PASS		
CORE-005	Lobby loads independently of all individual game modules.	PASS	API-CORE-001	BR-LOBBY-001
CORE-006	Top navigation exposes all games equally.	PASS		BR-LOBBY-001
CORE-007	Admin navigation is available without replacing game navigation.	PASS		
CORE-008	Game modules are loaded dynamically on the frontend.	PASS		
CORE-009	Backend game modules are registered through clean API routes.	PASS		
CORE-010	A failure in one frontend game module is contained to that game view when possible.	PASS		
CORE-011	All API responses use a consistent ok/data or ok/error envelope.	PASS	API-CORE-001	
CORE-012	The app exposes /api/v1 versioned routes.	PASS	API-CORE-001	
CORE-013	The server writes no-store cache headers for API responses.	PASS		
CORE-014	The app supports state reset for testing.	PASS		
CORE-015	The app preserves compact 1080p layout using panel-level scroll areas.	PASS		BR-LOBBY-001
CORE-016	The casino state endpoint returns games, players, recent history, and ledger.	PASS	API-CORE-001	
CORE-017	The app stores persistent data under data/.	PASS		
CORE-018	The app stores isolated per-game state under data/games/.	PASS		
CORE-019	The app migrates older v7/v8 data best-effort.	PASS		
CORE-020	The README documents run and test commands.	PASS		

8.9 Documentation

ID	Requirement	Status	API tests	Browser tests
DOC-001	A PDF requirements and validation document is generated for the release.	PASS		
DOC-002	Markdown copy of requirements document is included.	PASS		
DOC-003	Requirements are numbered with stable IDs.	PASS	API-ADMIN-001	
DOC-004	Each requirement lists module ownership.	PASS	API-ADMIN-001	
DOC-005	Each requirement lists implementation files.	PASS		
DOC-006	Each requirement lists validation status.	PASS	API-ADMIN-001	
DOC-007	Each requirement lists API tests where applicable.	PASS		
DOC-008	Each requirement lists browser tests where applicable.	PASS		
DOC-009	Module-specific revision numbers are documented.	PASS		
DOC-010	Architecture diagram is included in the PDF.	PASS		
DOC-011	API surface diagram is included in the PDF.	PASS		
DOC-012	Data/logging diagram is included in the PDF.	PASS		
DOC-013	Release notes are included in the package.	PASS		
DOC-014	Release notes summarize fixed regressions.	PASS		
DOC-015	Known limitations are documented when applicable.	PASS		

8.10 Keno

ID	Requirement	Status	API tests	Browser tests
KENO-001	Keno supports numbers 1 through 80.	PASS	API-KENO-001	
KENO-002	Keno allows selecting 1 to 20 spots.	PASS	API-KENO-001	
KENO-003	Keno draws 20 unique numbers.	PASS	API-KENO-001	
KENO-004	Keno uses explicit payable rows for 1-20 spots.	PASS	API-KENO-001	
KENO-005	Keno ticket purchase debits immediately.	PASS	API-KENO-001	
KENO-006	Keno ticket cancellation refunds before draw.	PASS	API-KENO-001	
KENO-007	Keno payout is based on spots, catches, and wager.	PASS	API-KENO-001	
KENO-008	Keno stores last draws.	PASS	API-KENO-001	
KENO-009	Keno UI has unique test IDs for every number.	PASS		BR-KENO-001
KENO-010	Keno UI highlights selected numbers.	PASS		BR-KENO-001
KENO-011	Keno UI highlights drawn numbers.	PASS		BR-KENO-001
KENO-012	Keno UI highlights catches.	PASS		BR-KENO-001
KENO-013	Keno UI displays payable.	PASS		BR-KENO-001
KENO-014	Keno UI displays drawn balls.	PASS		BR-KENO-001
KENO-015	Keno draw animation reveals balls.	PASS		BR-KENO-001

ID	Requirement	Status	API tests	Browser tests
KENO-016	Keno bots can buy quick-pick tickets.	PASS	API-KENO-001	BR-KENO-001
KENO-017	Keno bot strategies include 3/5/10/20 spot quick picks.	PASS	API-KENO-001	BR-KENO-001
KENO-018	Keno auto play repeats selected ticket.	PASS		BR-KENO-001
KENO-019	Keno speed control supports slow/medium/fast.	PASS		BR-KENO-001
KENO-020	Keno writes history rows.	PASS	API-KENO-001	BR-KENO-001
KENO-021	Keno remains fake-money only.	PASS		BR-KENO-001
KENO-022	Keno browser tests avoid ambiguous text selectors.	PASS		BR-KENO-001

8.11 Ledger

ID	Requirement	Status	API tests	Browser tests
LEDGER-001	Each player has a persistent fake-money balance.	PASS		
LEDGER-002	Human player exists by default.	PASS		
LEDGER-003	Three bot players exist by default.	PASS		
LEDGER-004	Add-money requires a positive amount.	PASS		
LEDGER-005	All betting debits go through the ledger service.	PASS	API-ROU-001	
LEDGER-006	All payout credits go through the ledger service.	PASS		
LEDGER-007	Ledger events include player, game, round, transaction type, amount, before, and after balance.	PASS		
LEDGER-008	Ledger is append-only JSONL.	PASS		
LEDGER-009	Insufficient funds rejects debit transactions.	PASS		
LEDGER-010	Roulette bet placement debits immediately.	PASS	API-ROU-001	
LEDGER-011	Roulette clear-before-spin credits refunds.	PASS	API-ROU-001	
LEDGER-012	Blackjack initial deal debits immediately.	PASS	API-BJ-001	
LEDGER-013	Blackjack double down debits separately.	PASS	API-BJ-001	
LEDGER-014	Blackjack split debits separately.	PASS	API-BJ-001	
LEDGER-015	Blackjack insurance debits separately.	PASS	API-BJ-001	
LEDGER-016	Baccarat bet placement debits immediately.	PASS	API-BAC-001	
LEDGER-017	Baccarat bet cancellation credits refunds.	PASS	API-BAC-001	
LEDGER-018	Keno ticket purchase debits immediately.	PASS	API-KENO-001	
LEDGER-019	Keno ticket cancellation credits refunds.	PASS	API-KENO-001	
LEDGER-020	Bingo card purchase debits immediately.	PASS	API-BINGO-001	
LEDGER-021	Bingo reset before calls credits refunds.	PASS	API-BINGO-001	
LEDGER-022	Slot spin debits spin cost before settlement.	PASS	API-SLOT-001	
LEDGER-023	All game settlement payouts credit after results are known.	PASS	API-ROU-001	
LEDGER-024	Ledger is visible in the admin console.	PASS		
LEDGER-025	Player balances are visible in the main UI.	PASS		

8.12 Logging

ID	Requirement	Status	API tests	Browser tests
LOG-001	Application log is written as JSONL.	PASS		
LOG-002	Error log is written as JSONL.	PASS		
LOG-003	Client/browser log is written as JSONL.	PASS		
LOG-004	API requests are logged with method and path.	PASS		
LOG-005	Unhandled API exceptions are logged.	PASS		
LOG-006	Client window errors are posted to the backend logger.	PASS		
LOG-007	Recent app logs are exposed through admin APIs.	PASS	API-ADMIN-001	
LOG-008	Recent error logs are exposed through admin APIs.	PASS	API-ADMIN-001	
LOG-009	Recent client logs are exposed through admin APIs.	PASS	API-ADMIN-001	
LOG-010	Logs are displayed in the /admin console.	PASS	API-ADMIN-001	BR-ADMIN-001
LOG-011	Test results are written under logs/test-runs/.	PASS	API-ADMIN-001	
LOG-012	Bingo abandoned sessions are logged.	PASS		
LOG-013	Roulette spin results are logged.	PASS		
LOG-014	Blackjack deals are logged.	PASS		
LOG-015	Baccarat coups are logged.	PASS		

8.13 Roulette

ID	Requirement	Status	API tests	Browser tests
ROU-001	Single-zero roulette mode supports 0 and 1-36.	PASS	API-ROU-001	
ROU-002	Double-zero roulette mode supports 0, 00, and 1-36.	PASS	API-ROU-001	
ROU-003	Wheel mode cannot change while open bets exist.	PASS	API-ROU-001	
ROU-004	Zero rule selector supports normal, la partage, and en prison.	PASS	API-ROU-001	
ROU-005	Straight-up bets are legal for all wheel numbers.	PASS	API-ROU-001	
ROU-006	Split bets are legal for adjacent table numbers.	PASS	API-ROU-001	
ROU-007	Zero split 0/00 is legal in double-zero mode.	PASS	API-ROU-001	
ROU-008	Street bets are legal for each table row.	PASS	API-ROU-001	
ROU-009	Corner bets are legal for all four-number intersections.	PASS	API-ROU-001	
ROU-010	Six-line bets are legal for adjacent row pairs.	PASS	API-ROU-001	
ROU-011	Zero trios are exposed where layout-appropriate.	PASS	API-ROU-001	
ROU-012	First-four/top-line bets are exposed by mode.	PASS	API-ROU-001	
ROU-013	Dozen bets are exposed and pay 2:1 net.	PASS	API-ROU-001	
ROU-014	Column bets are exposed and pay 2:1 net.	PASS	API-ROU-001	
ROU-015	Red and black outside bets are exposed.	PASS	API-ROU-001	
ROU-016	Odd and even outside bets are exposed.	PASS	API-ROU-001	

ID	Requirement	Status	API tests	Browser tests
ROU-017	Low and high outside bets are exposed.	PASS	API-ROU-001	
ROU-018	Snake bet is exposed.	PASS	API-ROU-001	
ROU-019	Racetrack call bets include Snake.	PASS	API-ROU-001	
ROU-020	Racetrack call bets include Voisins.	PASS	API-ROU-001	
ROU-021	Racetrack call bets include Tiers.	PASS	API-ROU-001	
ROU-022	Racetrack call bets include Orphelins.	PASS	API-ROU-001	
ROU-023	Racetrack call bets include Jeu Zero.	PASS	API-ROU-001	
ROU-024	Racetrack call bets include number and neighbors.	PASS	API-ROU-001	
ROU-025	Racetrack call bets include final digit.	PASS	API-ROU-001	
ROU-026	Racetrack call bets include complete number.	PASS	API-ROU-001	
ROU-027	Straight-up winning bets credit stake plus 35:1 profit.	PASS	API-ROU-001	
ROU-028	Split winning bets credit stake plus 17:1 profit.	PASS	API-ROU-001	
ROU-029	Street winning bets credit stake plus 11:1 profit.	PASS	API-ROU-001	
ROU-030	Corner winning bets credit stake plus 8:1 profit.	PASS	API-ROU-001	
ROU-031	Line winning bets credit stake plus 5:1 profit.	PASS	API-ROU-001	
ROU-032	La partage returns half of even-money outside bets on zero.	PASS	API-ROU-001	
ROU-033	En prison carries eligible even-money outside bets into the next round.	PASS	API-ROU-001	
ROU-034	Roulette spin writes history rows.	PASS	API-ROU-001	
ROU-035	Roulette stores last 1000 roll results.	PASS	API-ROU-001	
ROU-036	Roulette stats include frequency by number.	PASS	API-ROU-001	
ROU-037	Roulette stats include red/black/green counts.	PASS	API-ROU-001	
ROU-038	Roulette stats include odd/even counts.	PASS	API-ROU-001	
ROU-039	Roulette stats include low/high counts.	PASS	API-ROU-001	
ROU-040	Roulette stats include dozen and column counts.	PASS	API-ROU-001	
ROU-041	Roulette UI renders a vector wheel.	PASS		BR-ROU-001
ROU-042	Roulette UI animates wheel and ball while spinning.	PASS		BR-ROU-001
ROU-043	Roulette UI renders a real table-style layout.	PASS		BR-ROU-001
ROU-044	Roulette UI allows clicking number cells for straight bets.	PASS		BR-ROU-001
ROU-045	Roulette UI exposes clickable inside-bet spots.	PASS		BR-ROU-001
ROU-046	Roulette UI draws chips on table spots with amounts.	PASS		BR-ROU-001
ROU-047	Roulette auto play repeats saved bet templates.	PASS	API-ROU-001	BR-ROU-001
ROU-048	Roulette bots can be enabled with strategy and stake settings.	PASS	API-ROU-001	BR-ROU-001
ROU-049	Roulette scoreboard shows human and bot balances.	PASS		BR-ROU-001
ROU-050	Roulette voice announces the rolled number.	PASS		BR-ROU-001
ROU-051	Roulette wheel must not default to a fake zero result when no spin has occurred.	PASS		
ROU-052	Roulette wheel shows the latest actual spin result after settlement.	PASS		BR-ROU-001

ID	Requirement	Status	API tests	Browser tests
ROU-053	Roulette wheel selected pocket and backend spin result must match.	PASS		BR-ROU-001
ROU-054	Roulette spin animation starts before settlement display and ends on the result state.	PASS		
ROU-055	Roulette ball indicator uses the selected pocket when a result exists.	PASS		
ROU-056	Roulette sound settings are not rendered on the Roulette page.	PASS		

8.14 Slots

ID	Requirement	Status	API tests	Browser tests
SLOT-001	Slots uses five reels.	PASS	API-SLOT-001	
SLOT-002	Slots uses three visible rows.	PASS	API-SLOT-001	
SLOT-003	Slots uses reel-strip stop positions.	PASS	API-SLOT-001	
SLOT-004	Slots supports 1 payline.	PASS	API-SLOT-001	
SLOT-005	Slots supports 3 paylines.	PASS	API-SLOT-001	
SLOT-006	Slots supports 5 paylines.	PASS	API-SLOT-001	
SLOT-007	Slots supports 9 paylines.	PASS	API-SLOT-001	
SLOT-008	Slots supports 20 paylines.	PASS	API-SLOT-001	
SLOT-009	Slots evaluates wild substitutions.	PASS	API-SLOT-001	
SLOT-010	Slots evaluates scatter payouts.	PASS	API-SLOT-001	
SLOT-011	Slots awards free spins on scatter trigger.	PASS	API-SLOT-001	
SLOT-012	Slots carries free spins in persistent state.	PASS	API-SLOT-001	
SLOT-013	Slots maintains fake progressive jackpot.	PASS	API-SLOT-001	
SLOT-014	Slots resets progressive after jackpot hit.	PASS	API-SLOT-001	
SLOT-015	Slots exposes payable in API.	PASS	API-SLOT-001	
SLOT-016	Slots displays payable in UI.	PASS	API-SLOT-001	
SLOT-017	Slots debits non-free spin cost immediately.	PASS	API-SLOT-001	
SLOT-018	Slots credits wins after spin evaluation.	PASS	API-SLOT-001	
SLOT-019	Slots logs spin history.	PASS	API-SLOT-001	
SLOT-020	Slots UI animates reels.	PASS		BR-SLOT-001
SLOT-021	Slots UI highlights winning symbols.	PASS		BR-SLOT-001
SLOT-022	Slots UI shows recent spins.	PASS		BR-SLOT-001
SLOT-023	Slots auto play repeats selected line setup.	PASS		BR-SLOT-001
SLOT-024	Slots speed control supports slow/medium/fast.	PASS		BR-SLOT-001
SLOT-025	Slots sounds play during spin.	PASS		BR-SLOT-001
SLOT-026	Slots remains fake-money only.	PASS		BR-SLOT-001

8.15 Testing

ID	Requirement	Status	API tests	Browser tests
TEST-001	API test harness starts local server on a free port.	PASS	API-ADMIN-001	
TEST-002	API test harness resets state before tests.	PASS	API-ADMIN-001	
TEST-003	API tests validate core and admin endpoints.	PASS		
TEST-004	API tests validate roulette debit/settlement/rebet/en-prison.	PASS		
TEST-005	API tests validate slots spin behavior.	PASS		
TEST-006	API tests validate blackjack active-round protections.	PASS		
TEST-007	API tests validate baccarat deal behavior.	PASS		
TEST-008	API tests validate Keno paytable and draw behavior.	PASS		
TEST-009	API tests validate Bingo refund and auto-win behavior.	PASS		
TEST-010	Browser test harness uses Playwright.	PASS		BR-ADMIN-001
TEST-011	Browser tests reset state before tests.	PASS		BR-ADMIN-001
TEST-012	Browser tests use stable data-testid selectors.	PASS		BR-ADMIN-001
TEST-013	Browser tests capture console errors.	PASS		BR-ADMIN-001
TEST-014	Browser tests capture page errors.	PASS		BR-ADMIN-001
TEST-015	Browser tests save screenshots on failure.	PASS		BR-ADMIN-001
TEST-016	Browser tests validate lobby load.	PASS		BR-ADMIN-001
TEST-017	Browser tests validate roulette wheel/table/chips/spin.	PASS		BR-ADMIN-001
TEST-018	Browser tests validate slots spin UI.	PASS		BR-ADMIN-001
TEST-019	Browser tests validate Keno number selectors.	PASS		BR-ADMIN-001
TEST-020	Browser tests validate Bingo card UI.	PASS		BR-ADMIN-001
TEST-021	Browser tests validate Blackjack hand UI.	PASS		BR-ADMIN-001
TEST-022	Browser tests validate Baccarat deal UI.	PASS		BR-ADMIN-001
TEST-023	Browser tests validate Admin UI.	PASS		BR-ADMIN-001
TEST-024	Test results are written to JSON for admin viewing.	PASS	API-ADMIN-001	BR-ADMIN-001

8.16 Tests

ID	Requirement	Status	API tests	Browser tests
TEST-025	Browser tests use stable data-testid selectors for autoplay and admin controls.	PASS		BR-AUTO-ROU-001, BR-AUDIO-001
TEST-026	Browser tests verify Roulette autoplay stop behavior.	PASS		BR-AUTO-ROU-001, BR-AUDIO-001
TEST-027	API tests verify bot controller endpoints.	PASS	API-CONTROL-001	
TEST-028	API tests verify persisted audio settings.	PASS	API-CONTROL-001	
TEST-029	API tests verify server-registered autoplay session lifecycle.	PASS	API-CONTROL-001	
TEST-030	Test results remain visible in Admin.	PASS		

8.17 UX

ID	Requirement	Status	API tests	Browser tests
UX-001	Game stages reserve fixed visual areas during normal gameplay.	PASS		BR-ROU-001
UX-002	Action rails remain stable while actions execute.	PASS		BR-ROU-001
UX-003	Result messages render in fixed-height regions.	PASS		BR-ROU-001
UX-004	Long history, stats, paytables, and logs use internal scroll areas.	PASS		
UX-005	Autoplay status changes do not resize the main game stage.	PASS		
UX-006	Animations prefer transform/opacity and avoid layout-changing motion.	PASS		